UMUT-II HYDROPOWER PLANT, TURKEY

Document Prepared by GTE Karbon Sürdürülebilir Enerji Eğt. Dan. ve Tic. A.Ş.

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

RHG Enertürk Enerji Üretim ve Ticaret A.Ş. operates Umut-II Hydropower Plant, Turkey with 24.450 MWe (25.305 MWm) run-of-river hydropower electricity plant (HEPP) is located on Karakus River, Ordu, Türkiye. The expected life time of the project is roughly 49 years 0 months which corresponds to the duration of the licence obtained from EMRA. Project is a grid connected renewable power generation project which adds electricity capacity from hydro power sources which otherwise would have been generated by the fossil fuel fired power plant.

Main units that are constructed within Umut-II diversion weir and HEPP Project are: diversion weir, conveyance tunnel, forebay, penstock and HEPP building. Umut-I Hydropower Plant, Turkey & Umut-II Hydropower Plant, Turkey & Umut-III Hydropower Plant, Turkey have the same electricity generation licence and connected to the same electricity meters. Their installed capacities are different, and is represented in Table 3.

In the first crediting period, 3 turbines are installed with capacity of 3x8.15 MWe (3x8.435 MWm) (total capacity 24.450 MWe)[1]. There are also 2 hydraulic generators are installed, which have rated capacity of 9056 kVA. The net electricity production from the plant is estimated to be 68,740[2] MWh per annum. Estimated CO_2 reduction is 37,883 tonnes annually while it is calculated as 19,286 tonnes/year for this monitoring period. The actual electricity generation is 210,015.91 MWh for the whole monitoring period, which is between 21/02/2014 – 20/02/2020. The emission reduction achieved during this monitoring period is 115,715 tCO_2e.

The project milestones are provided in the table below:

Table 1. Project implementation milestones

Milestone	Date
Generation License	01/05/2008
Environmental Impact Assessment Approval	17/12/2009

[1] Provisional Acceptance documents of Umut-II HEPP

[2] Registred PD of the project

Commissioning of the Project (Project Start Date)	21/02/2014
Environmental Impact Assessment Exemption	06/07/2017
First Crediting Period	21/02/2014 – 20/02/2024
First Monitoring Period	21/02/2014 - 20/02/2020

Table 2. Technical Properties of Umut-II Hydropower Plant, Turkey Project

Location	Karakus River, Ordu Province, TÜRKİYE
Total Installed Capacity	24.45 MWe
Number Of Units	3
Turbine Capacity	3 x (8.435 MWm / 8.150 MWe)
Turbine Type	3 x Francis turbines
Average Annual Power Generation	68,740 MWh

Table 3. Installed capacities of Umut I-II-III

Installed Capacity of Umut-I	2x3.037 MWm / 2x2.900 MWe = 6.074 MWm / 5.800 MWe
Installed Capacity of Umut-II	3x8.435 MWm / 3x8.150 MWe = 25.305 MWm / 24.450 MWe
Installed Capacity of Umut-III	3x4.255 MWm / 3x4.000 MWe = 12.765 MWm / 12.000 MWe

Audit Type	Period	Program	VVB Name	Number of years
Validation/ Registration	21-February-2014 – 20-February-2024 (1st crediting period)	VCS	TÜV Rheinland Ltd.	10
Verification	21-February-2014 – 20-February-2020 (1st monitoring period)	VCS	Earthood	6

Total	-	-	-	10

1.2 Sectoral Scope and Project Type

Project Category: I.D. “Grid connected renewable electricity generation”

Scope: As per UNFCCC definition of sectoral scopes for CDM projects, the Project Activity is included in the Sectoral Scope 1, category “Energy Industries - Renewable Sources”.1

This is a voluntary project, but it follows the CDM rules. The approved baseline and monitoring methodology for small scale project activities ACM0002 Version.12.1: “Consolidated methodology for grid-connected electricity generation from renewable sources” is applied.

Proposed project activity is categorized in the sectoral scope 1 “Energy Industry – Renewable -/Non-renewable Sources” according to the UNFCCC definition[3].

The project is not a grouped project.

1.3 Project Proponent

Organization name	RHG Enertürk Enerji Üretim ve Ticaret A.Ş.
Contact person	Kutalmış
Title	Karacan
Address	Huzur Mah. Cendere Cad. Skyland İstanbul B Bok Kat:16 No:114B/300 Sarıyer/İSTANBUL/TÜRKİYE
Telephone	+0 212 267 42 06
Email	kutalmis.karacan@enerturk.com

1.4 Other Entities Involved in the Project

Organization name	GTE Karbon Sürdürülebilir Enerji Eğt. Dan. ve Tic. A.Ş.

[3] http://cdm.unfccc.int/DOE/scopes.html

Role in the Project	Project consultant
Contact person	M. Kemal Demirkol
Title	Director
Address	Mustafa Kemal Mah. 2111. Sok. No: 5 06530 Cankaya - Ankara – TÜRKİYE
Telephone	+90 312 514 63 63
Email	kemal.demirkol@gte.com.tr

It is to be noted that The project consultant Mavi Surdurulebilir Kalkinma Proje ve Danismanlik Hizmetleri Ltd. Sti. mentioned in the registered PD as other entities involved in the project has changed to GTE Karbon Sürdürülebilir Enerji Eğitim Danişmanlik ve Ticaret A.Ş., as GTE was contracted for the first verificatiton period services.

1.5 Project Start Date

Project has started in 21/02/2014[4], which is the provisional acceptance for this project activity.

1.6 Project Crediting Period

Project's first crediting period includes the dates of 21/02/2014 – 20/02/2024 (both dates included). The crediting period is 10 years (renewable).

1.7 Project Location

The project is located in the Black Sea region of Türkiye. The Project lies across the borders of Bartin and Kastamonu Province, near Kurucasile Town and also has borders to Akkuş District in Ordu Province, in Türkiye. The diversion weir structure is built on Karakuş stream.

The location is close to Town of Akkus.

[4] Provisional acceptance

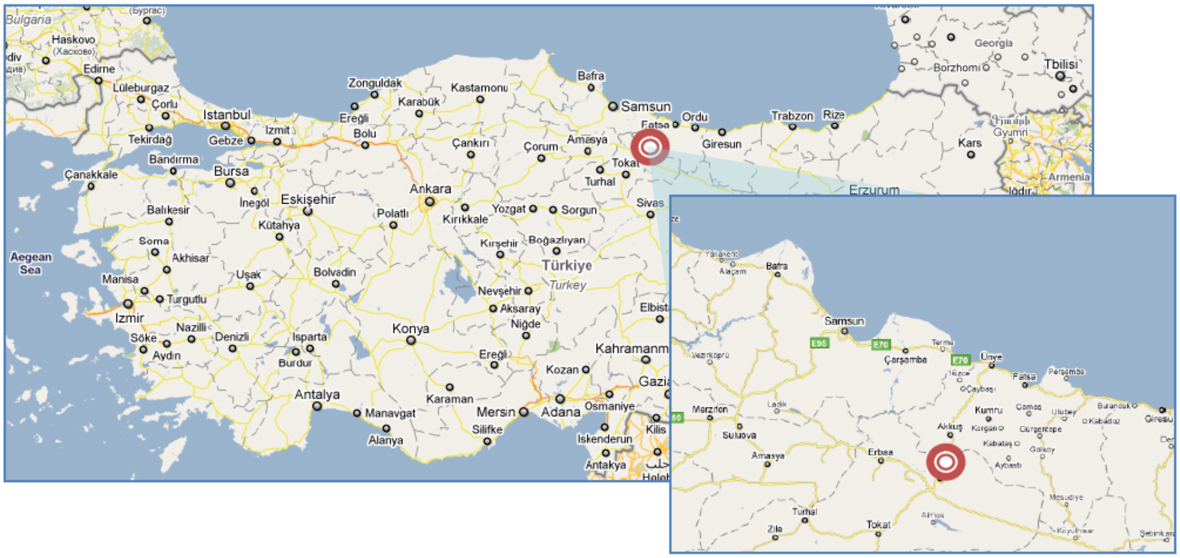

Figure 1. Project location

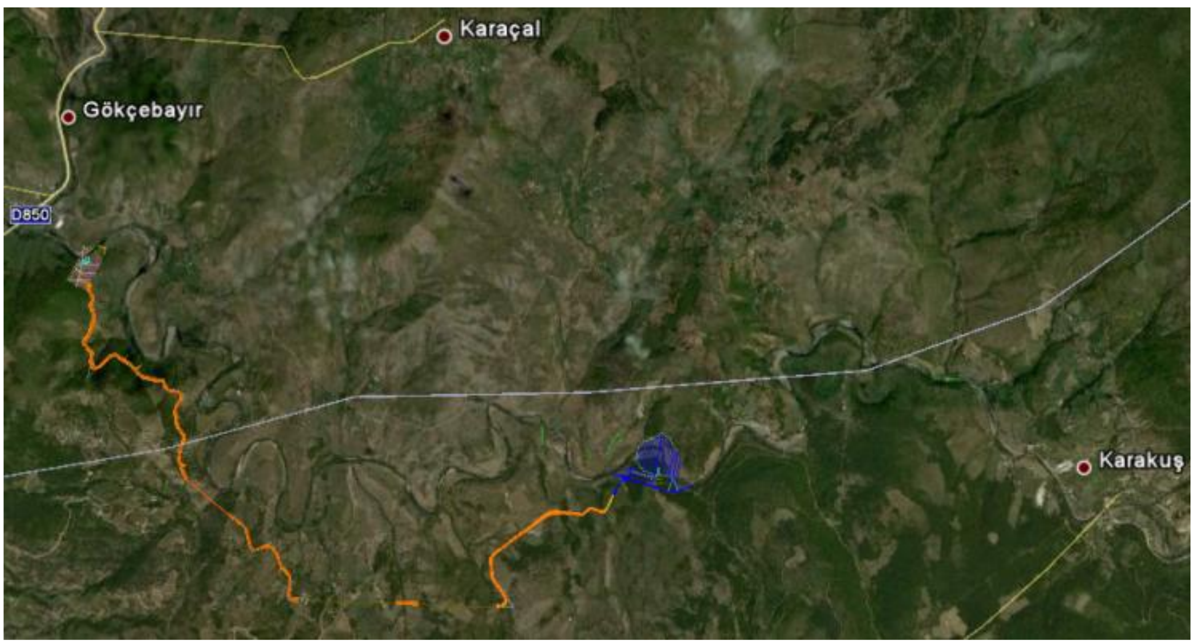

Figure 2. Location of the project on the map

GPS coordinates of the project area are given in the table below.

Table 4. Coordinates of the project area

Coordinate System	Diversion Weir	Power House
Degrees, Minutes, Seconds (Latitude)	40° 42’ 05”	40° 42' 46"
Degrees, Minutes, Seconds (Longitude)	37° 4’ 12”	37° 1' 44"
Decimal (Latitude)	40.701389	40.712778

Decimal (Longitude)	37.07	37.028889

1.8 Title and Reference of Methodology

According to the Appendix B of the simplified modalities and procedures for small-scale CDM-project activities, the methodology to be used for small scale renewable energy projects is ACM0002. 'Consolidated methodology for grid-connected electricity generation from renewable sources, Version 12.1, Scope 1, EB 48.

The UNFCCC approved baseline and monitoring methodology ACM0002, version 12.1 was applied for the project activity. Following tools have been used:

- The UNFCCC methodology "Tool to calculate the emission factor for an electricity system", Version 02.1.0.

- Tool: The CDM methodological additionality tool, "Tool for the demonstration and assessment of additionality", v05.2 is used for the assessment of additionality.

1.9 Participation under other GHG Programs

The project doesn't claim Green or White certificates or equivalents that may result in double counting as a result of carbon dioxide emission reduction purposes. The project does not claim Green or White certificates or equivalents that may result in double counting as a result of carbon dioxide emission reduction purposes.

1.10 Other Forms of Credit and Supply Chain (Scope 3) Emissions

Emission Trading Programs and Other Binding Limits: There is no other form of environmental credit generated by the project because there is no such system within the host country. The projects originate from Türkiye do not comply for renewable energy certificates of EU because there is no energy trade between EU and Türkiye because of different grid structures.

Other Forms of Environmental Credit: There is no other form of environmental credit generated by the project because there is no such system within the host country. The project does not generate other form of environmental credits such as Green Power Certificates. The projects originate from Türkiye do not comply for renewable energy certificates of EU because there is no energy trade between EU and Türkiye because of different grid structures.

The project does not claim Green or White certificates or equivalents that may result in double counting as a result of carbon dioxide emission reduction purposes. The Project has not created any form of environmental credit and the emission reductions of the Project have not been registered under any other programme. The Project does not intend to pursue any other form of GHG-related credits.

The VCU generated by the project activity will not be used in any other carbon offsetting program.

Supply Chain (Scope 3) Emissions: The project does not affects emissions associated with a good or service, demonstrate that a public statement(s) by the owner(s) or retailer(s) of the impacted good(s) or service(s) or project proponent (as applicable) has been made throughout the project crediting period.

1.11 Sustainable Development Contributions

The project helps following sustainable development goals:

1 – Affordable and Clean Energy (SDG 7): The project promotes clean and sustainable energy production and use. The total net electricity production of the project is 210,015.91 MWh for this monitoring period.

2 – Decent Work and Economic Growth (SDG 8): The project generates employment for all (12 employments for this monitoring period). The employees have required technical training and certifications.

3 – Climate Action (SDG 13): The project contributes to SDG 13 with an amount of 19,286 tonnes of CO_2e per annum for this monitoring period, which represent direct and quantifiable impact on climate security.

Table 5. Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	13.2	13.2.2 Total greenhouse gas emissions per year	Implemented activities to increase	By generating electricity from clean sources, project has prevented 115,715 tCO_2 into the atmosphere during the monitoring period	Expected prevented release of 115,715 tonnes of CO_2 into the atmosphere from the start of the project until the end of this crediting period.
2)	8.5	Job opportunities created	Implemented activities to increase	Due to the activity of the power plant, 12 people were employed.[5]	Due to the activity of the power plant, 12 people were employed during the monitoring period
3)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	The project generated 210,015.91 MWh electricity from hydropower which is a renewable source during the monitoring period	The project expected to generate 210,015.91 MWh electricity from hydropower which is a renewable source from the start of the project until the end of this crediting period.

[5] Social security records showing the employment.

2 SAFEGUARDS

2.1 No Net Harm

Since the project is a renewable energy project, it has positive impacts instead of negative impacts. The project uses hydropower to generate electricity rather than burning fossil fuels and causing pollution, hence, the project reduces emissions. The project provides employment opportunities as well.

Domestic and hazardous wastes (if any), and wastewater generated at the site will be handled according to the national regulations hence will not cause harm.

2.2 Local Stakeholder Consultation

An official public EIA meeting was not required by the Ministry. However, Project owner's personnel have visited several local communities and discussed the Project with stakeholders. Site visits and local meetings suggest that the local people believe that the Project is a positive contribution to their region.

Local people are mainly interested in fair compensation for their lands. There is minimal private land in the Project region, therefore there is limited amount of negotiations and no involuntary relocation of households. The stakeholders do not bring up any significant social or environmental highlight about the Project activity. There is no DNA in Türkiye. The National Focal Point of Türkiye is the Ministry of Environment and Forestry. The Turkish National Focal Point has been involved in the consultation process. Meetings with affected villages indicated that villagers are informed about the projects and potential impacts. Most frequently repeated questions and demands involve fair and timely land and housing compensation, employment opportunities for youth, and health and safety measures in particular to prevent drowning accidents. Since an EIA report was not required by the Ministry of Environment and Forest, no official EIA meeting was organized. Instead, various unofficial meetings have been carried out between the project owner and local communities. In addition, a stakeholder consultation meeting has been organized by project proponents on 10/04/2009 in Akkus town. Although the Project location is a remote location with no direct impact on surrounding villages, the closest communities are identified as local stakeholders and are invited. This meeting has been mentioned in the EIA report provided to the VVB. In this meeting, stakeholders have not provided any relevant positive or negative comment on the Project Activity. People are informed about the project and its impacts, which are primarily land acquisition and compensation. The project owner has committed to ongoing community engagement activities. Another secondary stakeholder feedback round was also organized in the region in 2010. No stakeholder provided any comments in this second stakeholder consultation that are relevant for the Project activity. Stakeholders did not report any critical issues regarding the Project activity. Therefore, there is no need to revise the Project.

Local stakeholders are able to communicate with the project site personnel verbally to share their grievances and inputs in a Continuous Input / Grievance Expression Process Book. A book is provided by PP is available with Chief of HPP to note down Input / Grievance Expression if any local stakeholder would like to share. The grievance might be negative as well as positive can be provided by local stakeholders related to operation of the project activity. PP will resolve all the grivances during whole crediting period. There were no grievances received during this monitoring period.

2.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project has been operational since 21/02/2014. Crediting period for the project is 10 years. The crediting period start date is 21/02/2014 and end date is 20/02/2024.

Main units that are constructed within Umut-II diversion weir and HEPP Project are: diversion weir, conveyance tunnel, forebay, penstock and HEPP building. Umut-I Hydropower Plant, Turkey & Umut-II Hydropower Plant, Turkey & Umut-III Hydropower Plant, Turkey have the same electricity generation licence and connected to the same electricity meters. Their installed capacities are different, and is represented in table below.

Table 6. Installed capacities of Umut HEPPs

Installed Capacity of Umut-I	2x3.037 MWm / 2x2.900 MWe = 6.074 MWm / 5.800 MWe
Installed Capacity of Umut-II	3x8.435 MWm / 3x8.150 MWe = 25.305 MWm / 24.450 MWe

Installed Capacity of Umut-III	3x4.255 MWm / 3x4.000 MWe = 12.765 MWm / 12.000 MWe

In the first crediting period, 3 turbines are installed with capacity of 3x8.15 MWe (3x8.435 MWm) (total capacity 24.450 MWe).

The details about the meters are given in the table below.

Table 7. Information about meters

	Main Meter (current)	**Spare Meter (current)**	**Main Meter (removed on 14/10/2020)**	**Spare Meter (removed on 14/10/2020)**
Brand	EMH	EMH	ACTARIS	ACTARIS
Type	LXQJ-XC	LXQJ-XC	SL 7000 761 A	SL 7000 761 A
Class	1S	1S	0.5 S	0.5 S
Serial No	9420180	9420181	65000694	65000695
Installation Date	14/10/2020	14/10/2020	13/12/2010	13/12/2010
Test Dates	23/07/2020	23/07/2020	18/07/2018 - 23/07/2020	18/07/2018 - 23/07/2020

There isn't any event or any situation that occurred during the monitoring period, which can change the applicability of the methodology.

3.2 Deviations

The deviations do not impact the appropriateness of the baseline scenario, additionality, or applicability of the methodology.

3.2.1 Methodology Deviations

N/A. Electricity generation of Umut-II Hydropower Plant, Turkey is calculated in an unconventional method, which was not explained in detail in the PD, which has been explained in Section 1.1 and will be explained in Section 3.2.2 as well.

3.2.2 Project Description Deviations

Electricity production license dated 20/11/2008 was in name of "Nisan Elektromekanik Enerji Sanayi Ve Ticaret A.S". Then, Boydak Enerji Üretim Ticaret Anonim Şirketi purchased the project shares. After the amendment, the name was changed to Boydak Enerji Üretim Ticaret Anonim Şirketi. Then "Boydak Enerji Üretim Ticaret Anonim Şirketi" changed its name to "RHG Enertürk Enerji Üretim ve

Ticaret A.Ş.", which is the current licence owner's name. These changes are addresses in the Generation Licence issued by EMRA, in the amendment section (8th and 13th).

VCS has been contacted regarding the company's name change from "Boydak Enerji Üretim Ticaret Anonim Şirketi" to "RHG Enertürk Enerji Üretim ve Ticaret A.Ş.", which changed on 17/01/2022.

In the registered PD, "The Project is *expected* to start generating electricity as of 01.10.2012, which is the Project start date for this Project activity.". As per the commissioning document dated as 21/02/2014, the start date of the project is taken as 21/02/2014. The commissioning was after the registration. The evidence document has been provided to VVB. The dates in the registry system will be corrected after the submission.

A different method of metering gross electricity generation and electricity consumption has been applied by Umut-II HEPP. Instead of continuously metering gross electricity generated and electricity consumed by Umut-II HEPP, the meters only measured gross electricity generation of Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP combined. Only data available from the project field is the combined gross electricity production of Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP. This is due to the fact that Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP are both connected to the same grid connection point and only one meter is responsible to meter both plant's data.

The project owner states that a proportional constant is used to calculate gross electricity production of Umut-II HEPP alone and this constant can be determined by using the ratio between plant capacity of Umut-II HEPP and the total plant capacity of Umut HEPP. It is provided that project capacity of Umut-I HEPP, Umut-II HEPP, and Umut-III are 5.800 MW, 24.450 MW and 12.000 MW, respectively.

Hence, to calculate gross electricity production only of Umut-III HEPP, the combined gross electricity production of the these plants are multiplied by the constant: 24,450/(5,800+24,450+12,000) = 0.5787.

EPİAŞ values of the three plants are also combined. Both gross electricity generation and electricity consumption are provided by EPİAŞ. To find the portion that is due Umut-II HEPP, EPİAŞ values are also multiplied by the proportional constant explained above.

Summary of information provided by the project owner:

- Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP are connected to the same grid connection point and the metering is done with one meter.

- Gross electricity generation data provided by the project owner shows three separate gross electricity production data set one belonging to Umut-I HEPP, one belonging to Umut-II HEPP, and one belonging to Umut-III HEPP. This does not indicate that the three different plants were metered separately. But instead, the total value was rationed between Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP considering their plant capacities.
- EPİAŞ data stands for values for Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP combined.
- Umut-I HEPP, Umut-II HEPP, and Umut-III HEPP operates under RHG Enertürk Enerji Üretim ve Ticaret A.Ş.

On the EPİAŞ and TEİAŞ system two values can be read: 1) Gross electricity generation 2) Internal electricity consumption. Transmission losses are already deduces from the gross electricity generation, therefore, no other action is required. Gross electricity generation is multiplied by the constant 0.5787 to calculate gross electricity generation by Umut-II HEPP in a month.

In the registered PD parameter table EGfacility,y the source of monitoring is "PMUM reports: PMUM is the financial settlement center of TEIAS, which has an online user interface for reporting purposes. PMUM web site can list a detailed overview of electricity generation and consumption figures of power plants, based completely on the measurement device readings carried out by TEIAS personnel." However, in 01/09/2015, market operations were moved from PMUM to EPİAŞ (Enerji Piyasaları İşletme A.Ş.). Therefore, PMUM database can not be accessed anymore. The electricity generation data has been transferred to EPİAŞ database. Screenshots sent by EPİAŞ are used for 2014, 2015, 2016, 2017, 2018, 2019, and 2020. As a cross-checking method, meter readings on-site (which are then sent to TEİAŞ) are used. The source of data are provided for each month in the monitoring period in the Emission Reduction Excel Spreadsheet.

PMUM has been replaced with EPİAŞ throughout the MR, compared with the registered PD.

3.3 Grouped Projects

The project is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	Official emissions from power generation

Data unit	tCO_2
Description	National GHG Inventory of Türkiye submitted to UNFCCC
. Source of data	UNFCCC web site[6]
Value applied	Used in: Table 12 in the Appendix
Justification of choice of data or description of measurement methods and procedures applied	These are 2007-2009 official emissions data submitted to UNFCCC for power generation facilities in Türkiye.
Purpose of the data	Baseline emission calculation
Comments	-

Data / Parameter	$GEN_{i,y}$
Data unit	MWh
Description	The gross electricity generation by fuel type j in year y (2005-2009) by the system
Source of data	TEIAS web site[7]
Value applied	Used in: Table 11 in the Appendix The sum of of $GEN_{j,y}$ for each fuel type in year y gives the System Gross Electricity Generation$_y$.
Justification of choice of data or description of measurement methods and procedures applied	TEIAS annually publishes official data regarding electricity generation and is the only source that provides information with most details.
Purpose of the data	Baseline emission calculation
Comments	-

[6] Source: Worksheet 1s1 in the http://unfccc.int/files/national_reports/annex_i_ghg_inventories/national_inventories_submissions/application/zip/tur-2011-crf-13apr.zip

[7] https://www.teias.gov.tr/turkiye-elektrik-uretim-iletim-istatistikleri

Data / Parameter	System Net Electricity Generation$_y$
Data unit	MWh
Description	The difference between the total quantity of electricity generated by power plants/units and the auxiliary electricity consumption of power plants/units.
Source of data	TEIAS web site[8]
Value applied	Used in: Table 11 in the Appendix
Justification of choice of data or description of measurement methods and procedures applied	TEIAS annually publishes official data regarding electricity generation and is the only source that provides information with most details.
Purpose of the data	Baseline emission calculation
Comments	-

Data / Parameter	Net Delivery Ratio$_y$
Data unit	%
Description	The ratio of the System Net Electricity Generation to the System Gross Electricity Generation in year y.
Source of data	Calculation: By dividing System Net Electricity Generationy by System Gross Electricity Generation$_y$
Value applied	Used in: Table 13 in the Appendix
Justification of choice of data or description of measurement methods and procedures applied	Electricity delivered to the grid by each power plant/unit is unavailable. The Net Delivery Ratio$_y$ is used for approximating the net electricity amount delivered to the grid by power plants/units except lc-mr sources.
Purpose of the data	Baseline emission calculation

[8] https://www.teias.gov.tr/turkiye-elektrik-uretim-iletim-istatistikleri

Comments	This is a conservative assumption, since in general thermal power plants consume more energy for auxilaries than e.g. hydro plants.

Data / Parameter	GEN_y
Data unit	MWh
Description	The system net electricity delivered in year y by the total of all power plants in the grid, excluding lc-mr sources.
Source of data	Calculation: 1. Gross electricity generation excluding lc-mr sources = Total gross electricity generation from all sources - total gross electricity generation from lc-mr sources 2. Net electricity generation excluding lc-mr sources = Gross electricity generation except lc-mr sources * Net Delivery Ratio 3. Net electricity delivered to the grid except lcmr sources = Net electricity generation except lc-mr sources + Imports
Value applied	Used in: Table 13 in the Appendix
Justification of choice of data or description of measurement methods and procedures applied	Net electricity delivered to the grid by each power plant/unit or by fuel type is unavailable. Therefore this parameter is approximated by calculations.
Purpose of the data	Baseline emission calculation
Comments	-

Data / Parameter	$EF_{CO2,i,y}$
Data unit	tCO_2/TJ
Description	CO_2 emission factor of fossil fuel type i in year y
Source of data	The lower limits of the 95% confidence interval stated in the “2006 IPCC Guidelines for National Greenhouse Gas Inventories”, Volume 2, Chapter 1 (energy) Table 1.4.[9]
Value applied	Used in: Table 13 in the Appendix
Justification of choice of data or	This is in line with the “Tool to calculate the emission factor for an electricity system“.

[9] https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf

description of measurement methods and procedures applied	
Purpose of the data	Baseline emission calculation
Comments	-

Data / Parameter	Electricity $Import_y$
Data unit	GWh
Description	Net electricity imports delivered to the grid in year y
Source of data	TEIAS website [10]
Value applied	Used in: Table 11 in the Appendix
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of the data	To calculate the quantity of net electricity delivered to the grid
Comments	This is the total electricity imported and delivered to the national grid from connected electricity systems from neighbour countries.

Data / Parameter	$\eta_{m,y}$
Data unit	%
Description	Average net energy conversion efficiency of power unit m in year y
Source of data	UNFCCC methodological tool "Tool to calculate the emission factor for an electricity system" Annex I[11]

[10] Source: http://www.teias.gov.tr/istatistik2009/30(84-09).xls

[11] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v2.pdf

Value applied	Used in: Table 13 in the Appendix
Justification of choice of data or description of measurement methods and procedures applied	Default efficiency figures represent best available techniques and should be considered to be conservative.
Purpose of the data	To calculate the quantity of net electricity delivered to the grid
Comments	There are no official efficiency values available for recent power plants based on each power plant or each fuel type. Therefore default values are assumed.

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh
Description	Net electricity supplied to the grid by the Project activity in year y
Source of data	Primary source: PMUM web site Secondary source: invoices and TEIAS protocols
Description of measurement methods and procedures to be applied	Measurement will be carried out by accurate electricity meters. The collected data will be, once approved by TEIAS, used to calculate the net electricity supplied by the Project.
Frequency of monitoring/recording	Continuous monitoring and recording. Monthly reporting
Value applied	210,015.91 MWh

<table>
<tr><th>Monitoring equipment</th><td>

	Main Meter (current)	Spare Meter (current)	Main Meter (removed on 14/10/2020)	Spare Meter (removed on 14/10/2020)
Brand	EMH	EMH	ACTARIS	ACTARIS
Type	LXQJ-XC	LXQJ-XC	SL 7000 761 A	SL 7000 761 A
Class	1S	1S	0.5 S	0.5 S
Serial No	9420180	9420181	65000694	65000695
Installation Date	14/10/2020	14/10/2020	13/12/2010	13/12/2010
Test Dates	23/07/2020	23/07/2020	18/07/2018 - 23/07/2020	18/07/2018 - 23/07/2020

</td></tr>
<tr><th>QA/QC procedures to be applied</th><td>

There will be no sampling and all measured data will be used for monitoring. Measurement equipment will be calibrated and their accuracy will be ensured by TEAIS.

Both measurement devices will be approved by TEIAS, and the calibration will also be managed by TEIAS. According to the “Communiqué for Metering Devices to be used in the Electricity Market”, Article 9, paragraph b, the periodic calibration shall be made every 10 years[12]. Both the primary and secondary devices are sealed during first operation of the plant. In case there is a significant inconsistency (assessed by TEIAS) between two devices, the meters will be calibrated by TEIAS before the 10 years period. The Project owner can also request calibration of the meter(s). The calibration stations are inspected by the central organization of the same Ministry.

Both metering instruments are sealed and only accessible by TEIAS personnel.

PMUM data is also of high quality, as the data is used for invoicing purposes and is accessible online by the Project owner, minimizing the risks of data corruption, data archiving, loss of information etc.

</td></tr>
</table>

[12] “Communiqué for Metering Devices to be used in the Electricity Market”, Article 9, paragraph b.

	Generation figures from PMUM reports will be copied-pasted to the spreadsheet to prevent potential typos.
Purpose of the data	To calculate the quantity of net electricity delivered to the grid
Calculation method	$EG_{Facility,y}=EG_{export,y}-EG_{import,y}$ (Annual electricity generation of the plant is calculated as the sum of the monthly generations obtained from EPIAS records. After deducting the amount of imported electricity from the generated electricity, the net electricity supplied to the grid is evaluated.) UMUT-I HYDROPOWER PLANT, Turkey & UMUT-II HYDROPOWER PLANT, Turkey & UMUT-III HYDROPOWER PLANT, Turkey are connected to the same electricity meters. Hence, their electricity generation data are read from one source. Their installed capacities are different, the ratio of their installed capacities are used to divide the electricity generation to each of the plants. It is not possible to monitor electricity generation values separately.Installed capacities from the generation licence are as follows. Installed capacity of UMUT-I HPP: 2x3.037 MWm / 2x2.900 MWe = 6.074 MWm / 5.800 MWe Installed capacity of UMUT-II HPP: 3x8.435 MWm / 3x8.150 MWe = 25.305 MWm / 24.450 MWe Installed capacity of UMUT-III HPP: 3x4.255 MWm / 3x4.000 MWe = 12.765 MWm / 12.000 MWe The electricity generations for each plant is calculated as follows $$Generation\ of\ UMUT\ I\ HYDROPOWER\ PLANT, TURKEY = \frac{Total\ generation\ value \times Installed\ capacity\ of\ UMUT\ I}{Total\ installed\ capacity}$$ $$Generation\ of\ UMUT\ II\ HYDROPOWER\ PLANT, TURKEY = \frac{Total\ generation\ value \times Installed\ capacity\ of\ UMUT\ II}{Total\ installed\ capacity}$$ $$Generation\ of\ UMUT\ III\ HYDROPOWER\ PLANT, TURKEY = \frac{Total\ generation\ value \times Installed\ capacity\ of\ UMUT\ III}{Total\ installed\ capacity}$$ Consider if the electricity generation in year x has been 124,894 MWh, read from the meter which all the plants are connected.

	$$Generation\ of\ UMUT\ I\ HYDROPOWER\ PLANT, TURKEY = 124{,}894\ MWh \times \frac{12.000\ MWe}{42.250\ MWe} = 17{,}145.21\ MWh$$ $$Generation\ of\ UMUT\ II\ HYDROPOWER\ PLANT, TURKEY = 124{,}894\ MWh \times \frac{24.450\ MWe}{42.250\ MWe} = 72{,}275.93\ MWh$$ $$Generation\ of\ UMUT\ III\ HYDROPOWER\ PLANT, TURKEY = 124{,}894\ MWh \times \frac{5.800\ MWe}{42.250\ MWe} = 35{,}472.85\ MWh$$ $$Total\ generation = 17{,}145.21\ MWh + 72{,}275.93\ MWh + 35{,}472.85\ MWh = 124{,}893.99\ MWh \approx 124{,}894\ MWh$$ During verification, “net generation value” will replace “total generation value”. Where, (Net Generation) = (Gross Generation) – (Electricity consumed from grid) Total generation value from the generation licence is only considered here for estimation purpose only.
Comments	-

4.3 Monitoring Plan

The purpose of the monitoring plan is to ensure that the monitoring and calculation of emission reductions of the proposed project within the crediting period is complete, consistent, clear and accurate.

As the Project boundary is defined as the national grid of Türkiye, the baseline emissions from electricity generation activities in Türkiye are calculated and monitored based on national official data. The leakage during crediting period will be negligibly small and will not be monitored, as fossil fuel consumption during construction and operation of the project activity is minimal. Hence, the monitoring plan involves the net electricity generation by the Project activity;

- The two measurement instruments are not accessible by the project participant or any other party except TEIAS. This prevents any intervention and assures the accuracy and quality of the measurements.
- The measurement instruments in broad terms give two types of data; the total gross electricity generated and the total electricity consumed by the Project activity. The difference of these

two data is the net electricity generated. Furthermore, TEIAS cuts a certain percentage of the generation to account for transmission losses. The net electricity generation, which is to be monitored and to be used for baseline emissions, is the net electricity generation, without the cut, which is the difference of gross electricity generation and self electricity consumption.

- At the end of each monitoring period, the data from the monthly meter readings will be added up to obtain the total monitoring period net electricity generation. This figure will be multiplied with the combined margin emission factor, which has been calculated ex-ante.

The first one is the amount of electricity fed into the grid. There is already an electricity meter installation, and these records provided the data for the monthly invoicing to TEIAS which form the basis for EPIAS data. EPIAS data is developed remotely by TEIAS and shows the monthly electricity generation of the company.

The second data to be monitored is the CO_2 emission reductions. The parameter is monitored by using data from EPIAS records and Meter reading records-OSF forms, and calculated by using Türkiye's calculated combined margin.

The third data to be monitored is the number of certificates issued/trainings provided by the project to the employees. These parameter will be monitored by checking training records (including H&S) & other certificates provided by the project owner.

The fourth data to be monitored is the installed capacity of the project activity. The data is monitored from the displays on the equipment which show the installed capacity at the time, together with the electricity generation of the facility. The installed capacity can be monitored from turbine labels and also the generation licence.

The fifth data to be monitored is the reservoir area for HEPP projects in general. It is monitored by maps and surveys.

The project is operated by RHG Enertürk Enerji Üretim ve Ticaret A.Ş. which ensures the overall site management in accordance with Turkish Laws and technology providers' guidelines.

VER Team Members are expected to include the following staff of the HEPP:

Plant Engineer: Responsible for the control of the electricity supplied to the grid and imported from the grid with TEIAS. The electricity measurements are made by TEIAS remotely. The plant engineer checks these electricity measurement records and reports to the Operations Manager of the plant.

Accounting Manager: Responsible for keeping data about power sales, invoicing and purchasing.

Operations Manager: the VER coordinator, responsible for developing, executing, analyzing and improving the VER Monitoring/Reporting Procedures.

GTE KARBON SURDURULEBILIR ENERJI EGT. DAN. VE TIC. A.S.: Responsible for emission reduction calculations, preparing monitoring report and periodical verification process.

In order to demonstrate the emission reduction, only required data is the net electricity delivered to the grid by the project activity. TEIAS has started to extract generation data from the facilities remotely through the EPIAS (previously PMUM system) which forms the basis for the billing.

During the monitoring period EPIAS readings and meter reading records (OSF forms) were used as the source of calculation of electricity generation and emission reductions. Meter reading protocols were compared for cross-check. Generation data collected during crediting period was submitted to GTE who is responsible for calculating the emission reduction subject to verification. Generation data were used to prepare monitoring report which was used to determine the vintage from the project activity. This report was submitted to the duly authorized and appointed Validation/Verification Body "VVB" before verification period.

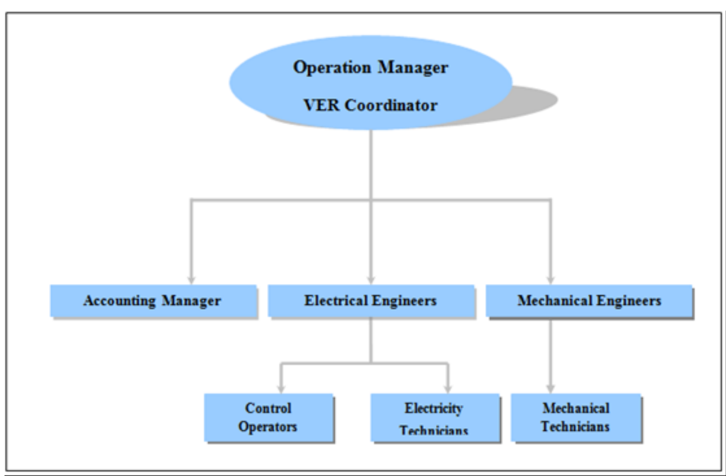

Figure 3. Site organizational chart

Each period, the monitoring report is submitted to DOE for the verification. The report covers the monitoring of grid-connected power generation, check report, report on calculation of the emission reductions and records of monitoring instrument repair and calibration, etc. All data collected as part of monitoring is archived electronically and kept at least for 2 years after the end of the last crediting period. The company establishes a dedicated maintenance system to ensure the data availability for the required period.

TEİAŞ is the responsible body for the meters and ensures the quality and accuracy of the measurements. Calibration of the meters are handled by TEİAŞ, hence no internal audit is performed by the power plant employees. Calibrations are done according to the national regulations by the distribution company. Calibrations are done according to the Measuring Instruments Directive[13]. The generation is zero for the period where the meters are being replaced. Meter replacements are done on the same day. Meters are tested every two years, according to the System Use Agreement between the PP and the distribution company TEİAŞ. Testing is done by TEİAŞ.

Local stakeholders are able to communicate with the project site personnel verbally to share their grievances and inputs in a Continuous Input / Grievance Expression Process Book. A book is provided by PP is available with Chief of HPP to note down Input / Grievance Expression if any local stakeholder would like to share. The grievance might be negative as well as positive can be provided by local stakeholders related to operation of the project activity. PP will resolve all the grivances during whole crediting period.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emission is calculated according to the formula

$$BE_y = EG_y \times EF_y$$

Where:

BE_y = Baseline emissions in year y (tCO_2)

EG_y = Net electricity delivered to the grid by the project activity in year y excluding transmission losses of the grid (MWh)

EF_y = Emission factor calculated according to selected methodology (tCO_2/MWh)

[13] https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5

$$210{,}015.91\ MWh \times 0.551\ tCO_2/MWh = 115{,}715\ tCO_2$$

$$BE_y = 115{,}715\ tCO_2$$

5.2 Project Emissions

As per ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources v.12.1.0.

For most renewable power generation project activities, PEy = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for, by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y = Project emissions in year y (tCO_2e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO_2/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO_2e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO_2e/yr)

- Since this project uses hydroelectric power, $PE_{GP,y}$ is "0" (tCO_2/yr)
- The only emission source in the plant is the diesel generator which is used as auxiliary power source when there is no electricity generation in the plant or supply by the grid. In urgent cases, a diesel powered generator will be operated for the daily consumption of personnel and the building, which is negligible. Project emissions are taken as zero. The same is also stated in para 33 of "ACM0002: Grid-connected electricity generation from renewable sources" ver 12.1. Hence it is neglected. Therefore; $PE_{FF,y}$ = 0 (t CO_2/yr)

Beside of the diesel generator, other potential project emission for this proposed project is $PE_{HP,y}$

However, if the power density of the project activity is greater than 10 W/m^2 than $PE_{HP,y}$ =0

The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity (W/m^2)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m^2)

A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m^2). For new reservoirs, this value is zero

For proposed project HEPP,

Cap_{PJ} = 24,450,000 W

Cap_{BL} = 0.0 W

A_{PJ} = 113,696 m^2

A_{BL} = 0.0 (m^2)

Therefore, PD is calculated as;

$$PD = \frac{(24{,}450{,}000\ - 0)\mathrm{W}}{(113{,}696\ - 0)\ \mathrm{m2}}$$

$$PD = 215.05\ W/m^2$$

Since the power density of the project activity is greater than 10 W/m^2, $PE_{HP,y}$ = 0.

5.3 Leakage

The energy generating equipment is not transferred from or to another activity. Therefore, leakage is also considered as “0”, according to the ACM0002 “Grid-connected electricity generation from renewable sources” methodology, version 12.1.

$$LE_y = 0$$

5.4 Net GHG Emission Reductions and Removals

Total Emission Reduction has been determined as;

$$ER_y = BE_y – PE_y – LE_y$$

Where;

ER_y = Emission reductions in year y (tCO_2)

BE_y = Baseline emissions in year y (tCO_2)

PE_y = Project Emissions in year y (tCO_2)

LE_y = Leakage emissions in year y (tCO_2)

The project emissions and leakage are considered as "0". Thus, $ER_y = BE_y$

Table 8. Calculations for net total values

EG_y	**Net Generation (MWh) During Monitoring Period**	**210,015.91**
EF_y	**Emission Factor (tCO_2/MWh)**	**0.551**
BE_y	**Gross Total Emission Reduction (tCO2) During Monitoring Period**	**115,715**
PE_y	**Project Emissions (tCO_2)**	**0**
LE_y	**Leakage Emissions (tCO_2)**	**0**
ER_y	**Net Emission Reduction (tCO_2) During Monitoring Period**	**115,715**

Thus, the net emission reduction (in tonnes CO_2) in this monitoring period (21/02/2014 to 20/02/2020) is calculated as given in table below.

Table 9. Summary of Emission Reductions

Year	Baseline Emissions (tCO_2e)	Project Emissions (tCO_2e)	Leakage Emissions (tCO_2e)	Net GHG Emission Reductions or Removals (tCO_2e)
2014 (21/02/2014 – 31/12/2014)	12,637	0	0	12,637

Year	Baseline Emissions (tCO_2e)	Project Emissions (tCO_2e)	Leakage Emissions (tCO_2e)	Net GHG Emission Reductions or Removals (tCO_2e)
2015	19,937	0	0	19,937
2016	22,904	0	0	22,904
2017	20,266	0	0	20,266
2018	17,187	0	0	17,187
2019	21,372	0	0	21,372
2020 (01/01/2020 – 20/02/2020)	1,412	0	0	1,412
Total	**115,715**	**0**	**0**	**115,715**

Total emission reductions realized as 115,715 tCO_2 for this monitoring period (Table 9). When the estimated electricity generation figure of the power plant for each year in the validated VCS PD (68,740 MWh/year) is considered, the total emission reductions should have been approximately 227,254 tCO_2 for the monitoring period (73 months). Percent difference is calculated as 49% which means the project reduced 49% less CO_2 than the estimated amount.

The produced electricity is fed into the Turkish National Grid through a long electricity transmission line. Hence, the long distance of the transmission lines causes differences in measured electricity. Therefore, the electricity generation values which are approved by EPİAŞ, are taken as the monitoring source.

Table 10. Summary of net electricity supply to the grid versus emissions reductions (estimate and actual values for this monitoring period)

Year	Project Baseline Estimate		Actual Values Achieved in the Monitoring Period	
	Net Electricity Supplied to the Grid (MWh)	Emission Reductions (tCO_2e)	Net Electricity Supplied to the Grid (MWh)	Emission Reductions (tCO_2e)
2014 (21/02/2014 – 31/12/2014)	59,135,23	32,584	22,936.45	12,637

Year	Project Baseline Estimate		Actual Values Achieved in the Monitoring Period	
	Net Electricity Supplied to the Grid (MWh)	Emission Reductions (tCO_2e)	Net Electricity Supplied to the Grid (MWh)	Emission Reductions (tCO_2e)
2015	68,740	37,876	36,184.72	19,937
2016	68,740	37,876	41,569.77	22,904
2017	68,740	37,876	36,781.07	20,266
2018	68,740	37,876	31,192.53	17,187
2019	68,740	37,876	38,787.98	21,372
2020 (01/01/2020 – 20/02/2020)	9,604.77	5,292	2,563.39	1,412
Total	**412,440**	**227,254**	**210,015.91**	**115,715**

Ex-ante emissions reductions/removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
227,254	115,715	-49	Since the project is a HEPP, seasonal variations in the amount of precipitation and temperature affect the monthly generation rates and may create deviations from the estimated values.

APPENDIX 1: RELEVANT DATA

Table 11. Gross and Net Electricity Generation [GWh] in Türkiye[14]

Generation DataBy Source, GWh	2005	2006	2007	2008	2009	Avg. Share
Coal	13,246	14,217	15,136	15,858	16,596	8.0%
Lignite	29,946	32,433	38,295	41,858	39,090	18.6%
Fuel Oil	5,121	4,232	6,470	7,209	4,440	3.4%
Diesel oil	3	58	13	266	346	0.0%
LPG	34	0	0	0	0	0.0%
Naphtha & Aphaltite	326	50	44	44	18	0.2%
Natural Gas	73,445	80,691	95,025	98,685	96,095	46.4%
Renewables and wastes	122	154	214	220	340	0.1%
Hydro	39,561	44,244	35,851	33,270	35,958	23.1%
Geothermal & Wind	153	221	511	1,009	1,931	0.2%
Gross Total	**161,956**	**176,300**	**191,558**	**198,418**	**194,813**	**100.0%**
Net Generation	155,469	169,543	183,340	189,762	186,619	
Net Delivery Ratio	96.0%	96.2%	95.7%	95.6%	96%	
Imports	635.9	573.2	864.3	789.4	812	
Gross Total excl. LC-MR	122,120	131,681	154,983	163,919	156,583	
Net Delivered to Grid, exc. lc-mr	117,864	**127,208**	**149,198**	**157,558**	**150,810**	

Table 12. Calculation of the OM emission factor[15]

Operating Margin Emission Factor	2007	2008	2009
Official CO_2 Emissions[16] [ktCO_2]	100,662	101,473	96,286
Net electricity delivered, excl. lc-mr [GWh]	149,198	157,558	150,810
Emission Factor [tCO_2/MWh]	0.675	0.644	0.638
Weight of generation, %	32.8%	33.9%	33.3%

[14] Source: TEIAS (http://www.teias.gov.tr/istatistik2009/32(75-09).xls , http://www.teias.gov.tr/istatistik2009/30(84-09).xls)

[15] Source: Worksheet 1s1 in the http://unfccc.int/files/national_reports/annex_i_ghg_inventories/national_inventories_submissions/application/zip/tur-2011-crf-13apr.zip

[16] Source: National GHG inventory report of Türkiye, as submitted to UNFCCC (http://unfccc.int/files/national_reports/annex_i_ghg_inventories/national_inventories_submissions/application/zip/tur-2011-crf-13apr.zip)

Operating Margin [tCO_2/MWh]	**0.653**

Table 13. Calculation of the BM emission factor

Capacity Additions	Total Generation [GWh]	Indicative Efficiency[17]	Emission Intensity [tCO_2/TJ][18]	Emissions [$ktCO_2$]	Build Margin [tCO_2/MWh]
Coal	1125.0	40%[19]	89.5	906.2	
Lignite	11455.8	40%[20]	90.9	9371.9	
Fuel Oil	2513.6	46%	75.5	1485.2	
Diesel Oil	0.0	46%	72.6	0.0	
LPG	0.0	60%	61.6	0.0	
Naphta	0.0	46%	69.3	0.0	
Natural Gas	18938.2	60%	54.3	6170.1	
Renewables	139.1		0	0.0	
Hydro	4962.3		0	0.0	
Geothermal & Wind	750.4		0	0.0	
Total	**39,884.3**			**17,933.4**	**0.450**

[17] Source: UNFCCC "Tool to calculate emission factor for an electricity system" , Annex 1

[18] Source: Lower values in the IPCC 2006 Guidelines V2, Ch1, Table 1.4

[19] The average design efficiency of coal power plants in Türkiye are 38%. Therefore assuming CFBS technology with 40% efficiency is appropriate and conservative. (Source: http://www.esmap.org/esmap/sites/esmap.org/files/Rpt_TürkiyeESM273.pdf)

[20] The average design efficiency of coal power plants in Türkiye are 38%. Therefore assuming CFBS technology with 40% efficiency is appropriate and conservative. (Source: http://www.esmap.org/esmap/sites/esmap.org/files/Rpt_TürkiyeESM273.pdf)

APPENDIX 2: SDG CONTRIBUTIONS

- **SDG 13: Climate Action and SDG 7: Affordable and Clean Energy**

T.C. ENERJİ PİYASASI DÜZENLEME KURUMU

ÜRETİM LİSANSI

Bu lisans kapsamındaki üretim tesisi Yenilenebilir Enerji Kaynağı kullanmaktadır.

(5346 Sayılı Kanunda yer alan, "Bu Kanun kapsamındaki yenilenebilir enerji kaynakları" tanımı çerçevesinde olan üretim tesisleri için)

Lisans No : EÜ/1590-1/1156

Tarih : 01/05/2008

Bu lisans, RHG Enertürk Enerji Üretim ve Ticaret Anonim Şirketi'ne, Ordu ve Tokat illerinde, Akkuş ve Niksar ilçelerinde kurulması planlanan Umut Reg. ve HES üretim tesisinde 01/05/2008 tarihinden itibaren 48 yıl, 8 ay, 4 gün süreyle, üretim faaliyeti göstermek üzere 4628 sayılı Elektrik Piyasası Kanunu ve ilgili mevzuat uyarınca Enerji Piyasası Düzenleme Kurulu'nun 01/05/2008 tarihli ve 1590-1 sayılı Kararı ile verilmiştir.

Mustafa YILMAZ
Başkan

EÜ/1590-1/1156

Figure 4. Cover page of the generation license

ÖZEL HÜKÜMLER

1- Üretim tesisine ilişkin bilgiler

Bu Lisans **RHG Enertürk Enerji Üretim ve Ticaret Anonim Şirketi**'ne ait ve bilgileri aşağıda yer alan **Umut Regülatörü ve HES** üretim tesisi için verilmiştir:

İli	: Ordu/Tokat
Bildirim adresi	: İkitelli Organize Sanayi Bölgesi Ziya Gökalp Mah. Atatürk Bulvarı 8. Cad No: 3 Başakşehir / İSTANBUL
Tesis tipi	: Yenilenebilir, hidrolik Rezervuarlı
Ünite sayısı	: 8 adet
Ünite kurulu güçleri	: (2x3.037)+(3x8,435)+(3x4,255)MWm/ (2x2.900)+(3x 8,150)+(3x4,000) MW_e
Tesis toplam kurulu gücü	: 44,144 MW_m / 42,250 MW_e
Enerji üretim kaynağı	: Hidrolik
Öngörülen ortalama yıllık üretim miktarı	: 124.894.000 kWh
Sisteme bağlantı noktası ve gerilim seviyeleri	: 154 kV Niksar HES Şaltina 154 kV Bolaman HES (Fatsa HES Projeleri) Şaltına 154 kV
Tesis tamamlanma tarihi	: 01/05/2014 İnşaat öncesi 30 ay İnşaat dönemi 42 ay

2- Lisansın yürürlüğe girmesi

Bu lisans, 01/05/2008 tarihinde yürürlüğe girer ve lisans sahibinin bu lisans kapsamındaki hak ve yükümlülükleri, lisansın yürürlük tarihinden itibaren geçerlilik kazanır.

3- Tüzel kişilikte yüzde on ve üzerinde doğrudan veya dolaylı pay sahibi olan gerçek ve tüzel kişiler

Doğrudan Pay Sahibi Ortak	Pay Oranı
İstikbal Mobilya San. ve Tic. A.Ş.	% 52,43
Bellona Mobilya San. ve Tic. A.Ş.	% 34,95

Dolaylı Pay Sahibi Ortak	Pay Oranı
Erciyes Anadolu Holding A.Ş	% 61,17
Huriye BOYDAK (Çocukları Hacı BOYDAK, Mustafa BOYDAK, Bekir BOYDAK, Memduh BOYDAK Nuran ŞENOZAN, Şükran BOZDAĞ ile birlikte)	% 33,39
Hacı Mustafa BOYDAK (Çocukları Şükrü BOYDAK ve Yusuf BOYDAK'la birlikte.)	% 24,41

4- Lisansın süresi

Bu lisans, yürürlük tarihinden itibaren 49 (kırkdokuz) yıl süreyle geçerlidir.

EÜ/1590-1/1156 1/5

Figure 5. First page of the generation license indicating the electricity generation value

- **SDG 8: Decent Work and Economic Growth**

T.C. ÇALIŞMA ve SOSYAL GÜVENLİK BAKANLIĞI
SOSYAL GÜVENLİK KURUMU BAŞKANLIĞI
SİGORTA PRİMLERİ GENEL MÜDÜRLÜĞÜ

SİGORTALI HİZMET LİSTESİ 20.10.2022 16:15:42

İşyeri Sicil No : [redacted]
İşyeri Ünvanı : RHG ENERTÜRK ENERJİ ÜRETİM VE TİCARET ANONİM ŞİRKETİ
SGM(kod-ad) : SGK ÜNYE SOSYAL GÜVENLİK MERKEZİ
İşyeri Adresi : UMUT HESÇAVDAR KÖYÜ Dış kapı no:0 İç kapı no:ORDUAKKUŞ
Yıl - Ay : 2022/08
Belge Çeşidi : 01
Mahiyet : ASIL
Kanun : 05510
Onay Tarihi : 23.09.2022
: [redacted]

Sno	S.Güvenlik No	Adı	Soyadı	İlkSoyadı	Ücret TL	İkramiye TL	Gün	UÇG	Eksik gün	Ggün	Çgün	İÇN	EGN	Meslek Kod
*1	[redacted]													3113.12
*2	[redacted]													3113.12
*3	[redacted]													3113.03
*4	[redacted]													3113.03
*5	[redacted]													3113.12
*6	[redacted]													3113.03
*7	[redacted]													3113.03
*8	[redacted]													7233.43
*9	[redacted]													9622.02
*10	[redacted]													3131.05
*11	[redacted]													3131.05
*12	[redacted]													7233.43
*13	[redacted]													3113.03
*14	[redacted]													2150.01
*15	[redacted]													3113.03
*16	[redacted]													3113.03
*17	[redacted]													3113.03
*18	[redacted]													9622.02
*19	[redacted]													9622.02
*20	[redacted]													3113.12
*21	[redacted]													3113.03
*22	[redacted]													3113.03
*23	[redacted]													2144.06
*24	[redacted]													3113.03
*25	[redacted]													3113.03
*26	[redacted]													3113.12

Figure 6. Social Security Records of Umut HEPP (Umut-I + Umut-II + Umut-III)

Santral	Personel Adı Soyadı	Görev Tanımı
Umut 1-2-3 HES	[redacted]	Santral Şefi
Umut 1-2-3 HES	[redacted]	Kıdemli Teknik Personel (Santral Sorumlusu)
Umut 1 HES	[redacted]	Kıdemli Teknik Personel
Umut 1 HES	[redacted]	Kıdemli Teknik Personel
Umut 1 HES	[redacted]	Kıdemli Teknik Personel
Umut 1 HES	[redacted]	Kıdemli Teknik Personel
Umut 1 HES	[redacted]	Teknik Personel
Umut 1 HES	[redacted]	Kıdemli Teknik Personel
Umut 1 HES	[redacted]	Genel Hizmet Elemanı
Umut 2 HES	[redacted]	Kıdemli Teknik Personel
Umut 2 HES	[redacted]	Teknik Personel
Umut 2 HES	[redacted]	Kıdemli Teknik Personel
Umut 2 HES	[redacted]	Kıdemli Teknik Personel
Umut 2 HES	[redacted]	Teknik Personel
Umut 2 HES	[redacted]	Teknik Personel
Umut 2 HES	[redacted]	Kıdemli Teknik Personel
Umut 2 HES	[redacted]	Kıdemli Teknik Personel
Umut 2 HES	[redacted]	Teknik Personel
Umut 2 HES	[redacted]	Genel Hizmet Elemanı
Umut 3 HES	[redacted]	Kıdemli Teknik Personel
Umut 3 HES	[redacted]	Kıdemli Teknik Personel
Umut 3 HES	[redacted]	Kıdemli Teknik Personel
Umut 3 HES	[redacted]	Kıdemli Teknik Personel
Umut 3 HES	[redacted]	Teknik Personel
Umut 3 HES	[redacted]	Kıdemli Teknik Personel
Umut 3 HES	[redacted]	Genel Hizmet Elemanı

Figure 7. Employee records indicating the number of employees working for Umut-II